

Case Study – 1: Enhancing Sales Performance with Excel Analytics

Background: Your company's leadership team is keen on identifying high-performing campaigns and optimizing regional strategies. As a Sales Analyst, your responsibility is to utilize Excel functions and tools to derive insights from the provided sales data. By leveraging functions such as VLOOKUP/XLOOKUP, IF, Pivot Tables, Conditional Formatting, Data Validation, and Charts, you will answer critical questions that will guide business decisions.

The dataset contains financial data with the following columns:

1. **Segment:** Market segment (e.g., Government, Midmarket).
2. **Country:** Country of sales.
3. **Product:** Product name.
4. **Discount Band:** Discount category applied.
5. **Units Sold:** Quantity of units sold.
6. **Manufacturing Price:** Cost to produce the product.
7. **Sale Price:** Selling price per unit.
8. **Gross Sales:** Total sales before discounts.
9. **Discounts:** Discount value applied.
10. **Sales:** Final sales revenue after discounts.
11. **COGS:** Cost of goods sold.
12. **Profit:** Profit earned from sales.
13. **Date:** Date of transaction.
14. **Month Number:** Month of transaction (numeric).
15. **Month Name:** Month of transaction (text).
16. **Year:** Year of transaction.

Practice Questions: Your dataset contains some missing values and potential inconsistencies. Perform data cleaning to remove inconsistencies: Identify and remove any duplicate rows in the dataset.

1. VLOOKUP/XLOOKUP

- **Task:** Fetch region-specific or product-specific data.
- **Q1:** Find the sale price of the product "Carretera" in "Canada."

2. IF Function: Classify ROI

- **Task:** Categorize transactions based on ROI (High or Low).
- **Q1:** Write an IF formula to classify a transaction as "High ROI" if Profit/Sales > 0.3; otherwise, classify it as "Low ROI."
- **Q2:** Identify all "Low ROI" campaigns and count how many exist in the dataset using COUNTIF.

3. Pivot Tables: Summarize by Region or Segment

- **Task:** Use Pivot Tables to analyse sales performance.
- **Q1:** Create a Pivot Table to summarize total profit by "Country" and "Segment." Identify the top 3 countries by profit.
- **Q2:** Drill down into the "Government" segment and find which country contributed the most profit.

4. Conditional Formatting: Highlight Top Performers

- **Task:** Visualize key metrics using Conditional Formatting.
- **Q1:** Apply Conditional Formatting to highlight transactions with a profit exceeding \$15,000.
- **Q2:** Use color scales to represent the sales performance of each product visually.

5. Data Validation: Filter Campaigns

- **Task:** Create dropdown lists for filtering. **Bonus:** Use dependent dropdowns, where the second list changes based on the first (e.g., selecting a "Segment" filters the "Country" options).
- **Q1:** Set up a dropdown list to filter the dataset by "Product."
- **Q2:** Combine dropdowns for "Segment" and "Country" to dynamically filter specific transactions.

6. Charts: Visualize Trends

- **Task:** Create charts to represent data insights.
- **Q1:** Generate a line chart showing monthly sales trends for the top 3 products.